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Plan Review

1.0 OVERVIEW

The importance of the semiconductor IH professional being involved in the plan review process is critical in semiconductor processing. The dynamic pace of technological changes in this industry creates an on-going need for the evaluation of process, production, and facility changes. With the useful life expectancies of state-of-the-art semiconductor processing equipment being measured in the 1–3 year duration, a continuous succession of reviews are needed.^[1]

The primary areas of importance for the IH involved in plan review process can be grouped into four primary areas:

- Ventilation
- Continuous Gas Monitoring
- Chemical Agent Safety
- Physical Agent Safety

Each of these areas have been discussed extensively in this text. Some additional key areas warranting mention are discussed in the context of an overall plan review program.

Many of the larger semiconductor manufacturers have incorporated tightly-designed plan review processes that require review and written approval from the company's Environmental Health and Safety (EH&S) representatives. Typically, the company's facility department will act as the lead group for coordinating the plan review process. Depending on the

size and scope of the project, a team of architects, engineers, process engineers, fire protection specialists, hazardous materials experts, industrial hygienists, and in-house EH&S representatives may be assembled. Many companies have developed detailed design specifications for the common elements of any construction or modification project. These specifications will address standard fire and life safety issues that are driven by local fire, building, hazardous materials and environmental requirements in the municipality that the facility is located. Providing an in-depth discussion of the overall plan review process is beyond the scope of this chapter. Several key points that need to be addressed by the EH&S group, and by the industrial hygienist specifically, are:

1. Types, quantities, and characteristics of the *hazardous production materials* (HPM) that are intended to be used, including information on the toxicology, reproductive and cancer causing properties (California Proposition 65 listing, IARC and NTP ratings, etc.), compatibility hazards, and methods of storage, use, dispensing, collection, and waste storage. Any use of radioactive materials or the generation of physical agents such as radio frequency (RF), infrared (IR), ultraviolet (UV), noise, or x-rays must be clearly identified.

Another area that deserves close scrutiny is the use of certain glycol ether based chemicals in photolithography that are considered reproductive hazards. Based on the epidemiological study results from U.C. Davis, Johns Hopkins University, and University of Massachusetts, the semiconductor industry is actively phasing out the use of these materials, and close attention should be given to finding suitable alternatives.

2. Equipment specifications, including the “cut sheets.” These sheets are specific to the piece of equipment, and provide a detailed inventory of the equipment’s physical, process, facility/environmental (power—routine and emergency), water, temperature, sanitary and hazardous material drains, and local exhaust ventilation. They routinely identify whether the equipment requires scrubbed exhaust for acids, volatile organics, pyrophorics, toxics, oxidizers, etc. Other information that is pertinent includes

the equipment's potential for particulate generation and the required static pressure in the area the equipment is located. Finally, the interconnection of this piece of process equipment to other pieces of equipment or wafer transport systems must be closely evaluated.

3. System safety characteristics/features, including the need for guarding, interlocks on power, physical energy sources such as radio frequency, and hazardous material supply and exhaust systems; fail-safe power, exhaust, and computerized control systems; continuous monitoring systems for hazardous materials that may be released within the various areas housing the operator interface, equipment support chase, and support equipment/hazardous material dispense areas, including exhaust conditioning and scrubbed exhaust emission points. Robotics is an area that is growing exponentially in the newer state-of-the-art fab areas. Careful review must be performed for the range of motion or travel, de-energized power state, and guarding/access control.
4. Hazardous material containment design, including secondary containment enclosures, plumbing/piping, vaulting, transport and dispensing systems and carts. It should include the use of the largest volume container that will provide the required purity levels to minimize the amount and quantity of manual materials handling, container cleaning/recycling, and to allow the remote dispensing of chemicals and waste collection to well-designed enclosures and rooms that have the system safety characteristics noted above.

Armed with the information that has been provided, the industrial hygienist can then perform an in-depth review of the proposed change, and ensure, through the signature loop, that key EH&S issues are addressed. A plan review can be a tug of war process, pitting the Facilities department against the "users" of the area, and the support departments such as EH&S, Security, Purchasing, and Process Engineering groups. Significant responsibility is vested to the EH&S department through signature authority on the building or modification plans. This authority requires that a

comprehensive understanding of the regulatory, company specifications, and “good IH practices” be available in EH&S group.

Typically, the plans for a larger project come in extensive sets of drawings on the basic site or building layout, mechanical, plumbing, and electrical. The lead times for review are generally very tight, and the information that was identified above is only partially provided in the initial stages of a project. This translates into a very demanding task that requires a firm resolve in ensuring appropriate EH&S standards and regulations are met. The industrial hygienist will need to be extremely well versed in the technical aspects of codes, and craft inventive strategies for complying with local or company requirements and still following a cost effective model. This is probably the most demanding aspect of an IH’s role in the semiconductor industry. The liberal use of individuals with specialized expertise—consultants and in-house personnel—technically knowledgeable in code compliance, ventilation, hazardous materials, systems safety, and chemical distribution systems is usually required along with the ability to provide a quick turn-around time on the task.

1.1 Release of Equipment or Area

Many progressive semiconductor companies have a program to evaluate newly constructed or modified areas or equipment. These are typically called an equipment sign-off or release program, and preclude the release of the applicable area or equipment until a final checkout has been performed by the EH&S group or certification by a consultant or testing firm. Specifically, the actual safety features of the equipment and area are functionally tested. This testing will include the tripping of safety interlocks, fail-safe features, challenging the continuous gas monitoring systems to known concentrations of gas, measuring the LEV levels or face velocities, measuring noise, physical agent, and chemical exposure levels. In some cases, depending on the enforcement practices of the local regulatory officials, they may want to physically see the testing, get copies of equipment certifications for electrical certifications (e.g., NEC compliance), gas monitoring, or other code compliance issues. This sign-off process allows the industrial hygienist to generate “baseline” IH monitoring levels for the equipment or area that are invaluable down-line as the equipment and areas begin to change overtime. With the on-going modifications to the chemistries used in the fab manufacturing processes, changes in the ventilation systems, and variations in through-puts on the equipment, on-going IH monitoring and tracking of results is essential.

Industry-accepted consensus standards have recently been developed by the Semiconductor Equipment and Materials International (SEMI) through their SEMI S2-93, *Safety Guidelines for Semiconductor Manufacturing Equipment*. This requirement is an excellent source of information in an equipment or plan review process. Figure 9.1 identifies the primary sections in the SEMI S2-1993 standard.^[2] Another pertinent document is one developed by the research consortium SEMATECH titled *Guidelines for Equipment Reliability, Technology Transfer*.^[3] Both of these documents will provide good basic documentation on equipment parameters that should be considered in any equipment review program.

1.2 Decommissioning and Decontamination of Equipment or Area

The need for a thorough removal of hazardous materials, residues, and disconnecting power and local exhaust sources is another area that deserves mentioning. With the dynamic nature of the industry, the need for a comprehensive methodology for “closing an area or piece of equipment” is mandatory. Many local regulations require that a piece of equipment or area be brought back to a “safe” state, and that testing records be provided confirming residue levels and waste disposal practices. The industrial hygienist should not underestimate the complexity of trying to decommission a complex piece of semiconductor processing equipment such as an ion implanter, CVD unit, diffusion furnace, photolithographic unit (coater), or plasma etcher. Weeks of lead time are needed to allow for regulatory approval of the closure or decommissioning plans, performing the decommissioning and submitting final results for approval. It is not atypical for this process to take from one to three months, depending on the specifics of the project.

There are no generally accepted procedures for determining appropriate decontamination levels for work surfaces in industrial buildings. Therefore, the suggestions in this section should be viewed as general guidelines that may not meet local regulatory requirements which tend to vary dramatically from one location to another, and are also influenced by the particular contractor used for the project.

Below is a procedure used to determine the maximum allowable contamination levels for a semiconductor fab that was decommissioned in 1993. The assumption was the building would remain in standard industrial use (e.g., manufacturing, office areas) and would not be converted to uses that might require more stringent standards such as a cafeteria or day care center.

- 1. Purpose**
 - 2. Scope**
 - 3. Safety Philosophy**
 - 4. General Guidelines**
 - 5. Safety-Related Interlocks**
 - 6. Chemicals**
 - 7. Ionizing Radiation**
 - 8. Non-ionizing Radiation**
 - 9. Audio Noise**
 - 10. Ventilation and Exhaust**
 - 11. Electrical**
 - 12. Emergency Shutdown**
 - 13. Heated Chemical Baths**
 - 14. Ergonomics / Human Factors**
 - 15. Robotics and Automation**
 - 16. Hazard Warning**
 - 17. Earthquake Protection**
 - 18. Documentation**
 - 19. Fire Protection**
 - 20. Environmental**
- Appendix 1 - Terminology**
- Appendix 2 - Referenced Documents**
- Appendix 3 - Supplemental Information**

Figure 9.1. The primary sections in the SEMI S2-1993 standard.^[2]

Irrespective of initial contamination levels, all potentially contaminated surfaces inside the building should be thoroughly cleaned (e.g., steam cleaned, or surface washing at least three times with a cleaning solution, changing the cleaning solution between cleaning cycles, or other generally accepted cleaning methods). This will help meet the OSHA surface contamination regulation that occurs in several chemical-specific standards: “All surfaces shall be maintained as free as practicable of accumulations of [the contaminant].”

After the thorough cleaning, samples should be taken using standard industrial hygiene wipe sampling procedures (e.g., see Ch. 2, Sec. 3.3). The number of wipe samples taken should be based primarily on the IH’s professional judgment of the number of samples needed to characterize a particular area. However, AHERA sampling strategies used for asbestos, with modification, can be applied to the contaminants of concern.

Maximum Allowable Surface Contamination Levels for Manufacturing Areas. If the area in the building where the contamination occurred is going to remain manufacturing, the maximum contamination levels are:

Cadmium	50 µg/100 cm ²
Lead	500 µg/100 cm ²
Nickel	500 µg/100 cm ²
Chromium	500 µg/100 cm ²
Copper	10,000 µg/100 cm ²
Tin (inorganic)	20,000 µg/100 cm ²

The following outline shows how the maximum contamination levels for the manufacturing areas were derived. The procedure assumes a potential only for ingestion of the contaminant, but it is based on dose associated with airborne occupational exposure limits. For additional information on wipe sampling, see Ch 2, Sec. 4.3.

1. Take the airborne standard, either the OSHA *permissible exposure limit* or the ACGIH *threshold limit value*[®] for the particular compound whichever is less:
 - Cadmium, OSHA PEL, 0.005 mg/m³
 - Lead, OSHA PEL, 0.05 mg/m³
 - Nickel, ACGIH TLV, 0.05 µg/m³

- Chromium, ACGIH TLV, 0.05 mg/m³
 - Copper, ACGIH TLV, 1 mg/m³
 - Tin (inorganic), ACGIH TLV, 2 mg/m³
2. Use the average amount of air inhaled during an 8-hr shift (10 m³) and assume total retention of the inhaled material during the shift, calculate an effective dose (e.g., 50 µg for cadmium, 500 µg for lead);
 3. Apply it to the typical surface area that is wiped to obtain the sample, 100 cm², to obtain the “standard” for the material (e.g., 50 µg/100 cm² for cadmium, 500 µg/100 cm² for lead).

While this method has some shortcomings in logic—people over the age of two do not lick work surfaces—it has credibility with some regulators (e.g., Fed-OSHA compliance officers which use it at least partially to determine sufficiency of work surface clean-up); it is also used by some contractors who specialize in the decommissioning of semiconductor fabs to determine sufficiency of decontamination.

This procedure is also the basis of a CAL-OSHA wipe sample requirement in their 4,4'-Methylenebis(2-Chloroaniline), MBOCA, standard. Section 5215(c)(2)(C) of the standard states: “Accessible surfaces which employees are likely to contact shall not exceed 100 µg MBOCA per 100 cm² of surface. For regulatory purposes, the average of at least five wipe samples shall be used; if one sample exceeds the mean by a factor of 10 or more, it will be rejected as not being representative.” MBOCA has a PEL of 10 µg/m³.

Maximum Allowable Surface Contamination Levels for Office Areas. If the area in the building where the contamination occurred is going to be converted to other industrial uses such as office areas, the maximum contamination levels are:

Cadmium	10 µg/100 cm ²
Lead	30 µg/100 cm ²
Nickel	200 µg/100 cm ²
Chromium	200 µg/100 cm ²
Copper	2,600 µg/100 cm ²
Tin (inorganic)	Not Available

The outline below shows how the maximum contamination levels for office areas were derived. It is based on the dose associated with EPA drinking water standards.

1. Take the *maximum contamination levels* for drinking water for the particular compound (40 CFR 141.62 and 141.80):
 - Cadmium, EPA Drinking Water MCL, 0.005 mg/l
 - Lead, EPA Drinking Water MCL, 0.015 mg/l
 - Nickel, EPA Drinking Water MCL, 0.1 mg/l
 - Chromium, EPA Drinking Water MCL, 0.1 mg/l
 - Copper, EPA Drinking Water MCL, 1.3 mg/l
 - Tin, (no established EPA Drinking Water MCL)
2. Use the amount of water consumption the EPA assumed when they established their drinking water standards, two liters per day, and calculate an effective dose (e.g., 10 µg for cadmium, 30 µg for lead).
3. Apply it to the typical surface area that is wiped to obtain the sample, 100 cm², to obtain the “standard” for the material (e.g., 10 µg/100 cm² for cadmium, 30 µg/100 cm² for lead).

As with the TLV[®]/PEL-based method for manufacturing areas, the glaring shortcoming in this approach is the surface area assumption used in calculating the dose, 100 cm². The authors are not aware of any basis for using 100 cm² other than it is the traditional area used for wipe samples. However, assuming a person ingests all removable contamination from 100 cm² of work surface seems fairly conservative, particularly when dealing with potential exposures to adults—a person may drink 2 liters of water a day but is unlikely to ingest materials from 100 cm² of surface area a day.

As can be seen from this brief overview, the plan review process is a comprehensive program that can and will pay major dividends from the upfront recertification of EH&S hazards prior to turning the equipment on, and then preventing problems after the equipment is turned off.

REFERENCES

1. Winslow, W., III, *Uniform Fire Code—The Code Development Process and Hazardous Materials Revisions in 1994*, paper presented at the Semiconductor Safety Association Conference (April 1993) (Unpublished)
2. Semiconductor Equipment and Materials International, SEMI document S2-93, *Safety Guidelines for Semiconductor Manufacturing Equipment*, Semiconductor Equipment and Materials International, Mountain View, CA (1993)
3. SEMATECH: *Guidelines for Equipment Reliability, Technology Transfer* #92031014A-GEN, SEMATECH, Austin, TX (May 1992)